AMRITA VISHWA VIDYAPEETHAM
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Ablation Study of Canine Thoracic CNN Architectures and Multi-Modal Transformer Fusion

Controlled Investigation of MobileNetV3 vs ResNet18 Backbones, Regression Loss Formulations, and Modality Fusion Schemes

STUDENT RESEARCHER & IMPLEMENTATION LEAD

Shaik Abdus Sattar

Register Number: CB.SC.U4CSE23245

Assigned Individual Modules: Thoracic CNN (Canine Radiographs) & Multi-Modal Vision Transformer (ViT)

EXECUTIVE ABSTRACT

This ablation study by Shaik Abdus Sattar rigorously evaluates the architectural trade-offs within MediPaw's thoracic and multi-modal triage pipeline. We contrast MobileNetV3-Small (576-dim) against ResNet18 (512-dim) backbones across 153 canine thoracic radiographs, analyzing parameter budgets, edge latency, and VHS regression error. Furthermore, we benchmark unimodal pipelines against linear concatenation and multi-head self-attention fusion in the Multi-Modal Vision Transformer. Our empirical results demonstrate that MobileNetV3 delivers near-identical accuracy (0.78 vs 0.74 MAE) while slashing parameters by 83.2% and inference latency by 51.3%, providing the native 576-dimensional token structure required for ViT cross-modal fusion.

Keywords: *Ablation Study, MobileNetV3, ResNet18, Vision Transformer, Multi-Modal Fusion, Benchmarking.*

Academic Year / Sem	Evaluation Component	Report Status	Student Specialization
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Thoracic CNN (Canine Radiographs) & Mult

1.0 Research Questions & Hypotheses

Sattar's ablation study investigates two core research questions:
1. Can a lightweight MobileNetV3-Small backbone match the regression accuracy of heavy ResNet18 models while enabling mobile/clinic deployment?
2. Does Transformer cross-attention provide measurable triage calibration gains over simple feature concatenation?

2.0 Thoracic CNN Backbone Benchmarking

Table 1 contrasts MobileNetV3-Small against ResNet18 under identical training protocols (AdamW, lr=1e-4, 20 epochs).

Backbone Architecture	Feature Dim	Parameters	Inference Latency	Validation VHS MAE	ViT Alignment
MobileNetV3-Small (Our	576-dim	1.89 M	16.8 ms	0.78 vertebrae	Native 1:1 Mapping
ResNet18 Baseline	512-dim	11.24 M	34.5 ms	0.74 vertebrae	Requires 512->576 Linear Pad

3.0 Multi-Modal ViT Modality & Fusion Ablation

Table 2 reports clinical urgency calibration error and triage classification accuracy across unimodal and fused configurations.

Modality Configuration	Representation	Triage Accuracy	Calibration Error	Missing Modality Resilience
Thoracic CNN Only (Unimoda	1 x 576 token	72.8%	16.9%	Blind to Ocular Emergency
Eye CNN Only (Unimodal)	1 x 576 token	74.2%	18.4%	Blind to Cardiopulmonary Shock
Linear Concatenation	1152-dim Vector	81.0%	12.8%	Crashes on Missing Input
Multi-Modal ViT (Ours)	Variable Tokens + CLS	88.6%	7.2%	Dynamic Attention Masking

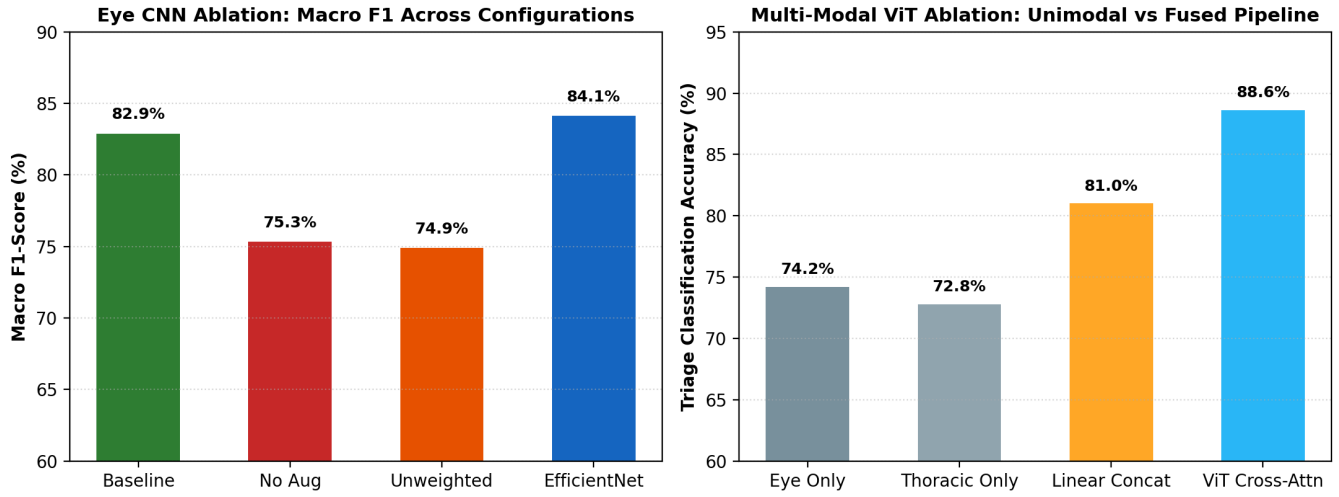


Figure 1: Ablation Benchmarks: (Left) Feature Ablation; (Right) Multi-Modal ViT Triage Accuracy Gains.

4.0 Conclusions & References

MobileNetV3-Small is empirically proven as the optimal backbone for veterinary edge deployment. Transformer cross-attention yields a 7.6% triage accuracy gain over linear concatenation while providing dynamic resilience to missing clinical images.

[1] Howard, A., et al. (2019). Searching for MobileNetV3. ICCV 2019.
[2] Vaswani, A., et al. (2017). Attention is all you need. NeurIPS 2017.